

Full Stack Intern (AI)

Build and ship a working SaaS product in 48 hours

DEADLINE	48 hours from the time this document reaches you.
AI TOOLS	Allowed and encouraged. Use whatever you like.
REQUIRED	A live, working, deployed website. Not a local project.

1 The Project

Freelancers and small studios still bill their clients with Word documents and spreadsheets. Invoice numbers get reused, totals are calculated by hand, and nobody can say quickly who has paid and who has not.

Build **BillFlow**, an invoicing product for freelancers. A user signs up, adds their clients, creates invoices with line items, sends them, and tracks what has been paid. Their client opens the invoice through a link and can see and pay it without creating an account.

2 Tech Stack

LAYER	USE
Frontend	Next.js with React and TypeScript, styled with Tailwind CSS
Backend	FastAPI with Python, or Next.js route handlers if you prefer to stay in one codebase
Database	PostgreSQL, with migration files
Hosting	Any provider you like, free tiers are fine

3 What to Build

1	Landing page	A public page that explains the product, with a sign up call to action.
2	Accounts	Sign up, log in, log out. Passwords hashed. A user only ever sees their own clients and invoices.
3	Clients	Add, edit, delete, and list clients. Name, email, company, address, phone.
4	Invoices	Create an invoice for a client with any number of line items, each with description, quantity, and rate. Totals, tax, and discount calculated automatically. Invoice number, issue date, due date, and notes. Status is draft, sent, paid, or overdue.
5	Invoice list	All invoices with search, filter by status and client, and sorting. Filtering happens on the server.
6	Invoice page	A clean, well designed invoice view that can be downloaded as a PDF or printed.

7	Sending	Send the invoice to the client by email, or generate a shareable link.
8	Public invoice	A page the client opens from that link without logging in, where they can view the invoice and pay it. A test mode payment or a simulated one is fine.
9	Dashboard	Total earned, outstanding, and overdue amounts, recent invoices, and a chart of income over time.
10	Settings	Business name, logo upload, currency, and invoice number prefix, all reflected on the invoice.
11	Overdue	An invoice past its due date shows as overdue without anyone marking it.
12	Everywhere	Loading, empty, and error states on every screen. Works on a phone. Looks like a real product.

4 Deliverables

ITEM	DETAIL
Live website	A deployed URL that works. Include demo login details so it can be opened and used immediately, and one invoice already created so the public invoice link can be seen.
GitHub repository	Public, or private with access given to the address in your invitation email.
README.md	What you built, the demo login details, setup steps to run it locally, and the environment variables. No real keys committed anywhere.
Migrations and seed	Migration files that build the database from empty, and a seed script with demo data.
Screen recording	Five minutes or less, with your voice, walking through the product you built.

5 Bonus

Extra credit for creativity and for features that make your version stand out. Anything you add beyond the list above, and any detail that makes the product feel finished, counts in your favour.

6 Submission

Reply to your invitation email before the deadline with the live URL, the repository link, and the recording. Where this brief does not specify something, decide for yourself and build it.